Development of a Stereo Deepwater Baited Remote Underwater Video System (DeepBRUVS)

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***Abstract*-Marine researchers aiming to acquire composition and size-frequency information on fish assesmblages have successfully used Baited Underwater Video Systems (BRUVS) as a non-extractive alternative to more traditional sampling methods using nets or traps. In a monitoring study of gulper shark populations on the eastern Australian upper continental slope (300-700 m depths), it was necessary to re-design the BRUVS lander to enhance its capabilities. The DeepBRUVS lander was designed to allow self contained and independent deep water operation (up to 1000m), with an extended deployment period of up to six months. DeepBRUVS is equipped with a stereo video system, lighting, and a bait release mechanism that permits multiple replicate video samples to be taken. The data will be used to provide relative abundance and size data as indicators of the viability of gulper shark populations protected within fishery closed areas. This paper discusses the design, prototyping and deployment of DeepBRUVS. The mechanical design of the bait piston, and camera assembly are reviewed in detail, along with a discussion of free fall and ascent dynamics. Bait types, bait release schedules, and camera recording sequences are also reviewed. The paper also discusses potential future additions and system improvements.**

I. INTRODUCTION

Baited Remote Underwater Video Systems (BRUVS) have a history of being successfully used to document the community composition and structure of fish assemblages in many shallow water applications [1-4]. BRUVS have some advantages over traditional net based sampling of fish assemblages including an ability to target a wide range of specific habitats (especially rocky reefs), and in being non-extractive.

Common net and trap-based sampling, are subject to selective sampling biases but these do not appear to compromise their utility to discriminate between assemblages or feeding types. For example, a comparison of BRUVS and prawn trawls [2] showed that while BRUVS recorded larger, mobile species from a much wider size range of families, both methods discriminated the same site groups. The use of bait raises questions about bias towards scavengers and predators, but a comparison of baited and unbaited BRUVS by Harvey et al. [3] showed that bait attracted greater numbers of predatory and scavenging species without decreasing the abundances of herbivorous or omnivorous fishes. That study concluded that a greater similarity between replicate samples from baited video within habitats implied the use of bait will provide better statistical power to detect spatial and temporal changes in the structure of fish assemblages and the relative abundances of individual species within them. BRUVS techniques have also been applied to study particular fish groups, and have provided good relative estimates of shark distribution and abundance. These methods have proven cost-effective for detecting rare species [5].

The successful use of baited cameras to study fishes in abyssal depths [6] has led BRUVS researchers to explore the potential of BRUVS configurations to work in deep continental shelf and continental slope depths off Australia and New Zealand. Within the last two years, deep water configurations with upgraded pressure casings have been used successfully in depths of at least 1,000 m [3].

BRUVS proven track record and its scope for deep water deployment led to the shallow water system being assessed for its ability to acquire monitoring data on demersal sharks in upper slope depths (300-700 m) off eastern Australia. The species of interest (gulper sharks from the genus Centrophorus) have been heavily depleted by fishing over much of their range and are being evaluated for threatened species listing [8]. The viability of remnant populations now protected in fishery closures will be monitored using relative abundance and distribution data. Non-lethal sampling is a pre-requisite for monitoring, while operationally the monitoring tool needs to be robust and deployable because costs-effective sampling will rely heavily on fishing vessels.

Initial trials with shallow BRUVS were promising, but they also revealed a number of necessary developments for the shark monitoring application. Principally, the need to collect data during commercial fishing operations will provide limited sampling time during infrequent survey opportunities. Acquiring sufficient sample numbers will therefore be dependent on multiple replicate video recordings at suitable intervals on long deployments, rather than numerous short deployments. Lights would be necessary to illuminate visiting sharks, and a mechanism was needed to generate bait plume to match interval recording. Lighting, video recording and the bait plume needed electronic control to synchronize the interval sampling. In addition, there were other ways to enhance the sampling program, including using a current meter to index bait plume dispersal, using acoustic releases to

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replace surface lines/floats, and deploying additional sensors to characterize the environment during long deployments. This paper describes these developments incorporated into the DeepBRUVS.

II. Conventional BRUVS

Traditionally, BRUVS systems consist of a simple frame to hold a camera, while a mesh bag with some fish bait would attract marine life in the surrounding area to the cameras field of view. These systems would typically be deployed in shallow water for short durations (on the order of hours or days) with a surface float for recovery. The cameras lacked external control and are enabled or disabled at the surface prior to deployment. As these systems are generally operated in shallow water, they would typically rely on natural lighting only. In some cases stereo camera systems have been deployed to enable measurement of fishes and have become increasingly popular. In order to attract marine life, fish bait (typically ground up pilchards) would be placed in a bag at the end of a boom in the line of sight of the cameras. This served to attract marine life in the surrounds until the bait was exhausted.

These simple systems have proven to be a reliable and cost effective way of collecting large amounts of video data of feeding marine life in shallow water. There are however some limitations owing to the simplicity of these systems. The lack of autonomous control over camera recording and bait release do not allow the system to operate for extended periods. In addition, the reliance on natural light combined with the lack of autonomous release mechanisms do not allow for use at depth. It was necessary to overcome these limitations in order to allow for the study of Gulper sharks in deep water.

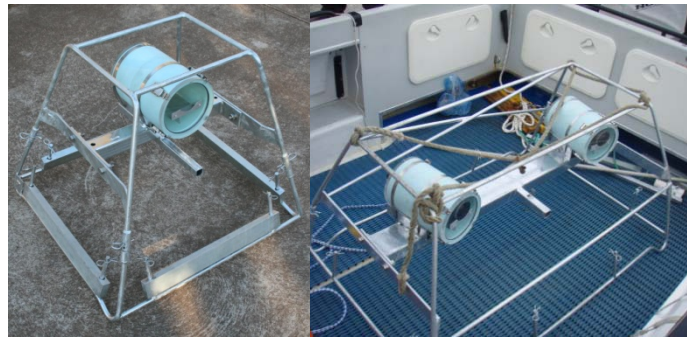

Figure 1. Conventional 2D(L) and 3D(R) BRUVS systems.

III. System Design

A. System Requirements

The study of Gulper sharks imposed some challenging requirements in the design of the DeepBRUVS system. The system would have to operate autonomously in depths up to 1000m for deployment periods of up to six months. Owing to the depth and deployment duration the systems were required to be designed as independent landers with no surface signature. Stereo video recording in high definition was required for sizing of targets, along with artificial lighting. A minimum of 24 hours of record time was imposed to allow scientists to make the most of each deployment. In order to attract targets on a regular and controlled basis, a bait

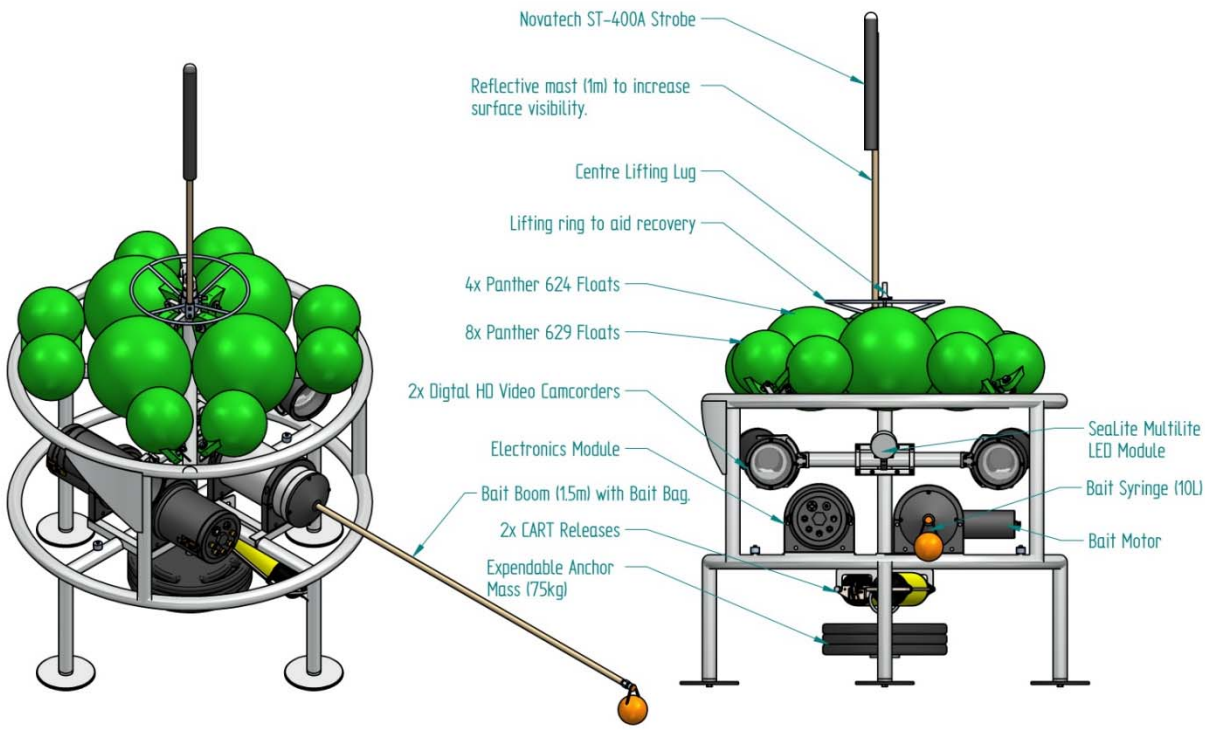

Figure 2. DeepBRUVS system concept.

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Figure 3. Front view of DeepBRUVS electronics bay.

Figure 4. Completed DeepBRUVS ready to deploy.

dispensing system would be required that would allow for the controlled release of 10 liters of liquefied bait. Finally, the system should have a small enough footprint to be deployed from a small vessel and an in air mass of less than 250kg. Once on the surface, the systems should have a turnaround time of less than 12 hours, to facilitate near continuous monitoring of sites. Finally the design should incorporate payload flexibility, such that additional oceanographic instruments such as conductivity temperature and depth probes (CTDs) and acoustic doppler current profilers (ADCPs) can be added in the future. After extensive consultation with science staff, a final design was approved and can be seen in figure 2.

B. *System Overview*

The adopted deepwater BRUVS design consists of a cylindrical style lander with a 1.25m diameter standing 1.4m tall. The landers frame is made of aluminum tubing and divided into three sections; floatation, instrumentation, and releases. The frame not only provides structural support for the equipment, but acts as a protective cage during deployment and recovery. Extending beyond the frame is a demarcation and positioning mast, and a bait release boom. The system is equipped with stereo high definition video cameras, a bait release piston, and electronics housing. Floatation is provided through the use of cost effective and readily available deepwater trawl floats. In this case four Panther 624 and eight Panther 629 floats were used. A fully loaded system weights 250kg in air, but only 15kg in water, with a reserve buoyancy of 60kg. There is sufficient reserve buoyancy that the system will still surface in the event of multiple float failure.

Systems are deployed at the surface using a davit or A-frame with a quick release mechanism. The system is designed to free fall to the desired depth without the aid of a drogue. Given the systems low weight in water, it descends at a leisurely rate of 0.5m/s taking around 30min to reach its maximum operating depth of 1000m. Four landing legs with pads distribute the system mass as it lands on the bottom. With a pad diameter of 0.2m the pressure on each pad is around 0.01 kg/cm^2 (or 0.16 psi). The hope is that the slow descent rate of the landers, combined with its minimal mass in water will reduce the probability of landers getting stuck in soft, muddy substrates.

Each system is equipped with a pair of ORE Offshore CART acoustic releases, which are connected to an expendable mass of steel weighing 75kg in air. Dumbbell weights have proven to be a convenient and cost effective anchoring solution for packages of this size. Once the mass is released the system returns to the surface at a more rapid rate of around 1m/s. A 1.5m reflective mast is used to increase the surface signature and aid in recovery. Systems are also equipped with a NovaTech strobe and radio range finder. A steel recovery ring allows the systems to be lassoed shipside for recovery. In order to provide good replication for gulper shark stock assessment, three identical landers were constructed.

C. *Camera System*

The BRUVS systems are each equipped with a pair of high definition video cameras. It was decided that commercial handheld cameras would be used as they are cost effective, and come complete with storage media. In addition, the cameras could easily be replaced as the technology improves. The systems are equipped with a pair of Panasonic HDC-HS700 cameras which have good low light performance, and high resolution 1080/50p recording capability. A 240GB storage capacity also allows for 21 hours of record time. Each camera is equipped with a wide angle Raynox DCR-6600Pro2 lens with an approximate magnification of 0.66x.

The cameras are housed in acetal housings with acrylic view ports. As the systems are being deployed for extended periods of up to 6 months, acetal was used to reduce corrosion on the housings. It was however necessary to pay close attention to creep and temperature affects in the design of the acetal housings. Internally, each camera is rigidly located and bolted in place with a locating ring which sits on the internal diameter of the housings. This is critical to maintaining stereo calibration stability throughout deployments. All camera control signals including USB are accessible via external

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Figure 5. Stereo video camera assembly.

connectors on each housing such that housings do not need to be opened during the survey. Data from both the video cameras and built in sensors can be downloaded through the external USB and serial ports respectively.

Pairs of camera housings are rigidly connected with an armature that fixes the cameras relative positions. This allows for easy separation of the camera assembly from the main DeepBRUVS landers for calibration and other general servicing. The aluminum armature is also sealed at both ends to add buoyancy. As a result, the camera assembly is neutrally buoyant at depth. On the armature, the cameras are spaced 0.6m apart at an angle of 8 degrees relative to centre. The focal point of each cameras field of view lies 2.8m from the camera lens, but owing to the use of wide angle lens adapters the cameras are able to have nearly full frame overlap a mere 1.5m from the housings. Once installed on the frame, cameras are trimmed to allow viewing of the sea floor at the 1.5m target distance. Lighting is provided by a single color balanced LED Multi-SeaLite Matrix lighting source by Deep Sea Power and Light. The Matrix lights have a maximum output of 2600 lumens, but also allow for power dimming, which can be adjusted to reduce target washout if needed.

Once assembled, cameras are calibrated both in air and in water to allow for target size measurement. Calibration is performed by recording motions of a calibration cube with reflective targets in two planes. Paired images are then processed in the computer and the cameras are paired. Typically, calibrations are performed before and after surveys to evaluate any calibration drift. As a result it is critical to ensure that the camera assembly is sufficiently rigid during normal use that the cameras are able to hold their calibration throughout a deployment.

Although cameras are calibrated, they are operated asynchronously, and so it is necessary to synchronize recorded frames during processing. This is accomplished using a simple array of LED's and digital counter, which is placed on the boom in the cameras field of view. Each frame will see a unique LED pattern and allow for frames to be synched during post processing.

D. Bait Release System

Extended BRUVS deployments require a bait dispensing system that will allow for the controlled release of bait, and hence controlled bait signal in the surrounding water. It was felt that the use of liquefied bait would provide the greatest flexibility in dispensing varying amounts of bait, and provide efficient dispersal of the bait plume. A large syringe with an internal diameter of 180mm and a volume of 10 liters was designed. The syringe is driven by a highly geared electric motor via a simple rack and pinion gear box. A boom at the end of the syringe releases the bait in the cameras field of view at a distance of 1.5m. A one way flow valve prevents the release of bait, and seals the piston between release periods. This allows for the removal of the bait signal, such that independent observations can be made over time. Fine motor control allows for bait to be released in 20cm^3 (0.68oz) increments.

Due to the nature of the piston and boom orifice size, only liquefied bait can be used. Non-perishable baits were required so that bait effectiveness would not decay during deployment. It was also vital to find bait that would disperse effectively in the surrounding water, maximizing its attractiveness. During the initial trials a mixture of fish oil and fish meals was used to assess system effectiveness.

E. Power, Electronics and Sensors

Each BRUVS system is powered by a single 53Ah rechargeable nickel metal hydride battery pack located in the electronics housing. The batteries are connected in seven strings and run at a nominal voltage of 16.8V. During operation all strings operate in parallel, but are charged independently to reduce heat build-up. Each pack is designed to provide a full 24 hours of stereo video recording, lighting and bait release.

A centrally located microprocessor controls all system components. Missions consisting of recording, lighting and bait release intervals can be pre-programmed externally using

Figure 6. Bait release piston.

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Figure 7. (L) Electronics housing connectors and interface. (R) Complete electronics housing with battery pack showing.

a laptop computer over an RS232 serial link. An external power switch allows for the system to be armed just prior to deployment, while an LED provides system status and health information. In addition to system control, the microprocessor records water temperature and depth. Camera video data however is stored locally on each camera and can be downloaded directly via externally accessible USB ports.

Initially, the systems have been configured to record the system descent and landing. Once on the bottom, the pre-scheduled science program will begin. The basic program releases bait and begins recording with lights on at various intervals. It is still unclear if this recording regime is the most effective at capturing targets, and the team is actively exploring additional configurations that may result in better target acquisition rates. Once the system has completed its pre-set instructions, or runs out of battery it shuts down. Throughout the mission, the system monitors its depth, and in the event of a premature system release will stop any current actions and power the system down for recovery.

IV. System Evaluation and Testing

Individual system components (cameras, electronics, bait release) systems were each tested on the bench to verify functionality. Following bench testing, a fully instrumented system was run in a controlled environment on land to assure proper system operation. A test from the CSIRO wharf in 10m of water was then conducted to trim system buoyancy and check the integrity of the video stream, and bait release mechanisms in water.

Once base functionality of the system was confirmed open water tests were conducted. These consisted of tethered and free deployments in around 100m of water to verify hydrodynamic stability of the DeepBRUVS platforms while descending and ascending. A three-axis self-logging accelerometer was installed to evaluate stability of the platform during deployment. Very short programmed test scenarios were also conducted to further test the video and electronics systems, as well as to replicate real world turnaround scenarios.

V. Conclusion and Future Work

The DeepBRUVS systems which were developed provide enhanced capabilities for monitoring gulper shark populations. Principally, they allow for deep water (up to 1000m) observations over extended periods (up to 6 months). The equipped stereo video system records high resolution stereo video of targets, allowing scientists to collect crucial size frequency information. The addition of a bait release system, allows for controlled bait signal and attraction of targets throughout the deployment period. Flexibility in the lander design allows for the addition of other instruments such as conductivity, temperature and depth sensors (CTDs) in the future. This will allow for the collection of a wide range of oceanographic data in situ with target size frequency information.

Future system improvements are currently being considered and tested in an attempt to improve performance. The current bait formula remains untested and further evaluation is needed to find a mixture that is most suited to attracting gulper sharks. The timing of video recording and bait release is being explored in hopes of increasing the footage time in which targets are in view. Active triggering mechanisms are also being considered to limit recording to times when targets are in view. This will allow for optimal use of battery power during extended deployments. Finally, it is envisioned that shark tag listening stations can be incorporated into the landers such that specific sharks can be identified in the video footage.

Acknowledgment

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References

[1] Willis, T.J., Babcock, R.C., 2000. A baited underwater video system for the determination of relative density of carnivorous reef fish. Mar. Freshw. Res. 51, 755– 763.

[2] Cappo M, Speare P, D'eath G (2004) Comparison of baited remote underwater video stations (BRUVS) and prawn (shrimp) trawls for assessments of fish biodiversity in interreefal areas of the Great Barrier Reef Marine Park. J Exp Mar Biol Ecol 302(2):123–152.

[3] Harvey ES, Cappo MC, Butler J, Hall N and Kendrick GA (2007) Bait attraction affects the performance of remote underwater video stations in assessment of demersal fish community structure. Marine Ecology Progress Series 350: 245-254.

[4] T. J. Langlois, E. S. Harvey, B. Fitzpatrick, J. J. Meeuwig, G. Shedrawi, D. L. Watson (2010). Cost-efficient sampling of fish assemblages: comparison of baited video stations and diver video transects. Aquatic Biology, 9: 155-168.

[5] Meekan, MG and Cappo, M. (2004) Non-destructive Techniques for Rapid Assessment of Shark Abundance in Northern Australia.

[6] Priede, I.G., Bagley, P.M., Smith, A., Creasey, S., Merrett, N.R., 1994. Scavenging deep demersal fishes of the Porcupine Seabight, north–east Atlantic: observations by baited camera, trap and trawl. J. Mar. Biol. Assoc. UK 74, 481– 498.

[7] Graham, KJ and Daley, RK (in press). Distribution, population structure and reproduction of gulper sharks (Centrophorus spp., Centrophoridae) in south-eastern Australian waters. Marine and Freshwater Research.

[8] Graham, K. J., Andrew, N. L., and Hodgson, K. E. (2001). Changes in relative abundance of sharks and rays on Australian South East Fishery trawl grounds after twenty years of fishing. Marine and Freshwater Research, 52: 549–61.

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